Book 1 — Lecture 2 — Exercise 4

Proving the Sum, Product, and Chain Rules

Prompt. Prove the sum rule (fairly easy), the product rule (easy if you know the trick), and the chain rule (fairly easy).[1]

Setup

Let f and g be differentiable at x.[2]

Sum rule

Define $S = f + g$. For $h \neq 0$,

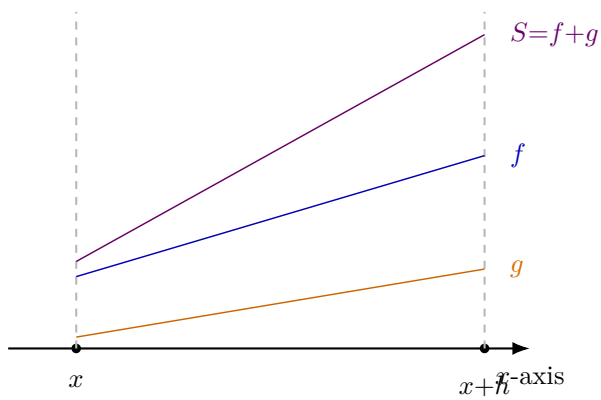
$$\frac{S(x+h) - S(x)}{h} = \frac{f(x+h) - f(x)}{h} + \frac{g(x+h) - g(x)}{h}.$$

Each quotient has a limit as $h \to 0$ by hypothesis; limits add termwise, so

$$S'(x) = f'(x) + g'(x).$$

Equivalently, $(f+g)' = f' + g'$ at each point where both derivatives exist.

Picture (sum). Over the *same* step from x to $x+h$, the vertical changes in f, g, and $S = f+g$ add: $\Delta S = \Delta f + \Delta g$. Dividing by h and letting $h \to 0$, the secant slopes add the same way the rises do.



Rises Δf, Δg over $[x, x+h]$ stack to ΔS; hence $\dfrac{\Delta S}{h} = \dfrac{\Delta f}{h} + \dfrac{\Delta g}{h}$.

[1] Throughout, work at a fixed point x in an open interval where the functions written are differentiable in the usual real sense. Limits are taken as $h \to 0$ with $h \neq 0$.

[2] The *derivative at* x is $f'(x) = \lim_{h \to 0} \dfrac{f(x+h) - f(x)}{h}$ when this limit exists.

Product rule

Write the Newton quotient for $P = fg$, then add and subtract the same term $f(x+h)g(x)$ in the numerator:[3]

$$\begin{aligned} P(x+h) - P(x) &= f(x+h)g(x+h) - f(x)g(x) \\ &= f(x+h)\big(g(x+h) - g(x)\big) + g(x)\big(f(x+h) - f(x)\big). \end{aligned}$$

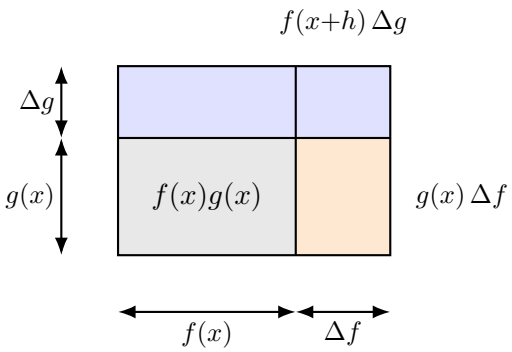
Divide by $h \neq 0$:

$$\frac{P(x+h) - P(x)}{h} = f(x+h)\,\frac{g(x+h) - g(x)}{h} + g(x)\,\frac{f(x+h) - f(x)}{h}.$$

Let $h \to 0$. By continuity of f at x, $f(x+h) \to f(x)$. Hence

$$P'(x) = f(x)g'(x) + g(x)f'(x).$$

Picture (product). Think of f and g as side lengths of a rectangle, so $P = fg$ is its area.[4] When f and g change, the new area splits into the old rectangle plus a "top strip" (width $f(x+h)$, height Δg) and a "right strip" (height $g(x)$, width Δf). That is exactly the two grouped terms before dividing by h.



Shaded: original area. Blue strip: $f(x{+}h)\big(g(x{+}h) - g(x)\big)$. Orange strip: $g(x)\big(f(x{+}h) - f(x)\big)$. Sum $= P(x{+}h) - P(x)$.

Chain rule

Let $F = f \circ g$ with $F(x) = f(g(x))$. Assume g is differentiable at x and f is differentiable at $u = g(x)$.[5]

For $h \neq 0$, define the "difference quotient for f along the increment of g" by

$$Q(h) = \begin{cases} \dfrac{f(g(x+h)) - f(g(x))}{g(x+h) - g(x)}, & g(x+h) \neq g(x), \\ f'(g(x)), & g(x+h) = g(x). \end{cases}$$

[3] This "trick" isolates increments of f holding g fixed at x, and increments of g holding a prefactor that becomes $f(x)$ as $h \to 0$ because differentiable functions are continuous.

[4] This is bookkeeping only: f and g are abstract quantities; the picture assumes h is small enough that both $f(x{+}h)$ and $g(x{+}h)$ lie on the same side of zero as $f(x)$ and $g(x)$ so lengths are positive.

[5] The composition need not be differentiable without both hypotheses at the matching points.

If $g(x+h) = g(x)$, then also $f(g(x+h)) - f(g(x)) = 0$, so $F(x+h) - F(x) = 0$ and

$$\frac{F(x+h) - F(x)}{h} = 0 = Q(h) \cdot \frac{g(x+h) - g(x)}{h}.$$

If $g(x+h) \neq g(x)$, then directly

$$\frac{F(x+h) - F(x)}{h} = Q(h) \cdot \frac{g(x+h) - g(x)}{h}.$$

So in either case the same product formula holds for all nonzero h.[6]

Because f is differentiable at $u = g(x)$, one has $Q(h) \to f'(u)$ as $h \to 0$ (when $g(x+h) \neq g(x)$, $Q(h)$ is the usual secant slope; when $g(x+h) = g(x)$ it is already pinned to $f'(u)$). Also $\dfrac{g(x+h) - g(x)}{h} \to g'(x)$. Products pass through limits, hence

$$F'(x) = \lim_{h \to 0} Q(h) \cdot \lim_{h \to 0} \frac{g(x+h) - g(x)}{h} = f'(g(x))\, g'(x).$$

Picture (chain). Two stages: an "inner" map $x \mapsto u = g(x)$, then an "outer" map $u \mapsto f(u)$. A small move h at x produces a small change $\Delta u \approx g'(x)\, h$ in the intermediate variable; that change, viewed on the u-axis, produces $\Delta F \approx f'(u)\, \Delta u$. Multiplying the local rates gives $\Delta F / h \approx f'(u) \cdot g'(x)$. The algebraic $Q(h)$ construction is the bookkeeping that makes the $\Delta u = 0$ case match the same formula.

$$x \longrightarrow \boxed{g} \xrightarrow[\Delta u]{} u{=}g(x) \longrightarrow \boxed{f} \xrightarrow[\Delta F]{} F{=}f(u)$$

inner slope $\dfrac{\Delta u}{h} \to g'(x)$ outer slope $\dfrac{\Delta F}{\Delta u} \to f'(u)$

Overall secant: $\dfrac{\Delta F}{h} = \dfrac{\Delta F}{\Delta u} \cdot \dfrac{\Delta u}{h}$ when $\Delta u \neq 0$; the piecewise $Q(h)$ extends this to $\Delta u = 0$.

Result. For differentiable f, g at x,

$$(f+g)'(x) = f'(x) + g'(x), \qquad (fg)'(x) = f'(x)g(x) + f(x)g'(x).$$

For $F = f \circ g$ with g differentiable at x and f differentiable at $g(x)$,

$$(f \circ g)'(x) = f'(g(x))\, g'(x).$$

Example. Sum: $(x^2 + \sin x)' = 2x + \cos x$. Product: $(x \sin x)' = \sin x + x \cos x$. Chain: $\dfrac{d}{dx} \sin(x^2) = 2x \cos(x^2)$.

[6] This unifies the $\Delta u = 0$ and $\Delta u \neq 0$ cases without separate limit arguments.